

Critical Bernoulli Site Percolation on Integer Lattices

A finite hyperedge inequality and a same-parameter slab argument

Research draft

Last updated: September 3, 2026

Verification warning. This manuscript presents a new proof. It has not yet been independently refereed or checked by a proof assistant. The parts requiring priority in an independent audit are the hyperedge extension of the conditioned covariance hierarchy in Sections 3–5 and the site exploration in Section 7. Compilation and finite enumeration are consistency checks, not substitutes for such an audit.

This research draft was produced by ChatGPT 5.6 with minimal human intervention.

Abstract

We prove that nearest-neighbour Bernoulli site percolation on every integer lattice has no infinite cluster at its critical parameter. The finite input is a multiplicative gluing inequality for independent Bernoulli hyperedges. Site percolation becomes such a model by replacing each open site with one open hyperedge joining its incident ports. The main difficulty is that two different hyperedges can reach the same boundary vertex, so ordinary boundary sets do not preserve intersections in the induction. We retain the hyperedge names as source labels and attach a vertex set to each label. Union and intersection are then exact at the label level, while connectivity still uses the union of the attached vertex sets. This gives the conditioned cluster inequalities needed for multiplicative gluing. A site version of the Kozma–Nitzan block construction then turns an infinite cluster at a parameter p into an infinite cluster in a finite-width slab at the same parameter. The half-space theorem for site percolation rules this out at criticality.

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1 The result

Let each vertex of $\mathbb{Z}^d$ be open independently with probability p . An open path is a nearest-neighbour path all of whose vertices are open. We write $C(x)$ for the open cluster of x , with $C(x) = \emptyset$ when x is closed. Define

$$\theta_{\text{site}}(p) = \mathbb{P}_p(C(0) \text{ is infinite}), \quad p_c^{\text{site}} = \inf\{p : \theta_{\text{site}}(p) > 0\}.$$

Theorem 1.1 (Critical site percolation). *For every integer $d \geq 2$,*

$$\theta_{\text{site}}(p_c^{\text{site}}) = 0.$$

The proof has two parts. First, Theorem 5.4 proves a finite gluing inequality for independent hyperedges. Corollary 6.1 transfers it to site percolation when the named terminals are forced open. Second, Proposition 7.8 inserts that corollary into a site-based block exploration. This produces slab percolation without increasing p . The final contradiction uses the half-space theorem of Barsky, Grimmett, and Newman.

The requirement that named terminals are open is natural and necessary. Without it, $\mathbb{P}(o \leftrightarrow A) \leq p_o$, so a near-one hypothesis is impossible when $p_o < 1$. In the lattice argument the source, target, and relays arise by wiring or by conditioning and are therefore open with probability one.

2 Independent hyperedges

We first describe the finite model. Let V be a finite vertex set, and let E be a finite set of labels. Each $e \in E$ has an incidence set $\iota(e) \subseteq V$ containing at least two vertices. Independently for each e , declare e open with probability $p_e \in [0, 1]$. Two vertices are connected when a chain of open hyperedges joins them. Connectivity is reflexive in this finite hypergraph model. For $S \subseteq V$, let C_S be the union of the connected components that meet S .

For conditional arguments we retain slightly more information. The **cluster record** of S is

$$\mathcal{K}_S = \{e \in E : e \text{ is open and } \iota(e) \cap C_S \neq \emptyset\}.$$

Its support is C_S , with the roots S retained even when no incident hyperedge is open. If $x \in S$, then C_x is recoverable from $\mathcal{K}_S$ by taking the component of x in the recorded open hyperedges. We order cluster records by inclusion. Their supports and each rooted component increase in this order.

For $U \subseteq V$, the induced hypergraph on U retains precisely the hyperedges whose incidence sets lie in U . If $K \subseteq U$, deleting K means passing to the induced hypergraph on $U \setminus K$; thus every hyperedge meeting K is removed. We use a superscript U when the ambient vertex set matters.

2.1 Site representation

The following exact construction explains why hyperedges are the right finite model. It also explains why replacing a site by independent incident edges would be incorrect.

Lemma 2.1 (Site-to-hyperedge representation). *Let $G = (V, E_G)$ be a finite graph with independent site probabilities $(p_v)_{v \in V}$. There is an independent hyperedge model such that, for all distinct vertices $x, y \in V$, the event $\{x \leftrightarrow y\}$ is identical in the two models. The identity also holds for $x = y$ when $p_x = 1$.*

Proof. For each graph edge $uv \in E_G$, introduce a new port m_{uv} . For each non-isolated site v , introduce the hyperedge

$$H_v = \{v\} \cup \{m_{uv} : uv \in E_G\}.$$

Declare H_v open with probability p_v , independently over v . No hyperedge is needed for an isolated site because it cannot participate in a connection between distinct vertices.

Adjacent sites u and v have intersecting hyperedges because $m_{uv} \in H_u \cap H_v$. It follows that an open site path gives a chain of open hyperedges. Conversely, consecutive hyperedges in a chain can intersect only at a port m_{uv} belonging to adjacent sites u and v . Removing the ports from the chain gives an open site path. This proves the event identity for distinct terminals. If $x = y$ and $p_x = 1$, then both models regard x as connected to itself. $\square$

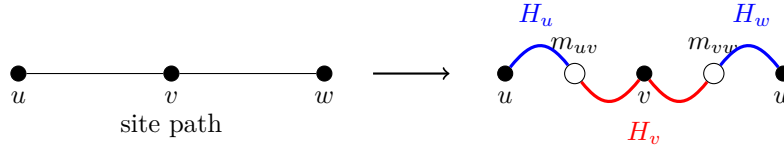


Figure 1: An open site activates one hyperedge containing the site and all of its incident ports. Adjacent open sites communicate through their common port. The colors distinguish labels; equal colors do not indicate a shared random variable.

2.2 Cluster exploration

Let $Z \subseteq U$. Write

$$E_Z = \{e \in E : \iota(e) \cap Z \neq \emptyset\}$$

for the hyperedges incident to Z . If $I \subseteq E_Z$ is the set of open incident hyperedges, define its trace outside Z by

$$T_Z(I) = \bigcup_{e \in I} (\iota(e) \setminus Z). \quad (1)$$

The trace obeys

$$T_Z(I \cup J) = T_Z(I) \cup T_Z(J), \quad T_Z(I \cap J) \subseteq T_Z(I) \cap T_Z(J). \quad (2)$$

The second relation need not be equality: two different hyperedges can reach the same vertex.

Lemma 2.2 (Exploration identity). *Let $N, Z \subseteq U$ with $Z \subseteq N$. Condition on the states of the hyperedges in E_Z , and let $I \subseteq E_Z$ be the open set. In every event on which a cluster from $U \setminus Z$ avoids N , deleting Z replaces the source N by*

$$(N \setminus Z) \cup T_Z(I).$$

The hyperedges disjoint from Z retain their original independent laws.

Proof. Every path that first enters Z uses an open member of E_Z . Its last vertex before leaving Z lies in the trace in (1). Conversely, each vertex in the trace is joined to Z by its defining open hyperedge. After the incident hyperedges have been exposed, the remaining coordinates are disjoint from E_Z and therefore keep their product law. These two observations give the claimed source replacement and the conditional law. $\square$

We also use exact conditioning on a cluster. If $C_S = K$, then each hyperedge meeting both K and $U \setminus K$ is closed. Hyperedges disjoint from K remain independent. Thus the unexplored configuration is precisely the original product model on the induced hypergraph $U \setminus K$.

3 Cluster inequalities

The finite gluing proof needs two forms of conditional association. We give the hyperedge proofs because the graph versions alone do not cover site percolation.

For an increasing nonnegative function f of a vertex cluster, write

$$K_U(f; X) = \mathbb{E}_U[f(C_s); C_s \cap X = \emptyset].$$

Lemma 3.1 (One-cluster inequality). *Let U be a finite vertex set in an independent hyperedge model. Let $s \in U$, let $X, Y \subseteq U$, and let f, g be increasing nonnegative functions of subsets of U . Then*

$$K_U(f; X)K_U(g; Y) \leq K_U(fg; X \cap Y)K_U(1; X \cup Y). \quad (3)$$

Consequently, conditional on $C_s \cap X = \emptyset$, increasing functions of C_s are positively associated.

Proof. We prove (3) by strong induction on $|U|$. Cases with $s \in X \cup Y$ have a zero factor. Let $Z = X \cap Y$.

Suppose first that $Z = \emptyset$. The maps $f(C_s)$ and $g(C_s)$ are increasing in the hyperedge coordinates, whereas both avoidance indicators are decreasing. Two applications of Harris' inequality, followed by Harris' inequality for $f(C_s), g(C_s)$ and for the two avoidance indicators, give (3).

For the inductive step, suppose that $Z \neq \emptyset$. Expose the states $I \subseteq E_Z$ and delete Z . Lemma 2.2 writes the four factors in (3) as sums over I . For two exposed states $I, J \subseteq E_Z$, apply the induction hypothesis in $U \setminus Z$ with the avoided sets

$$(X \setminus Z) \cup T_Z(I) \quad \text{and} \quad (Y \setminus Z) \cup T_Z(J).$$

The union of these sets equals

$$((X \cup Y) \setminus Z) \cup T_Z(I \cup J).$$

Their intersection contains $T_Z(I \cap J)$ by (2). Avoidance probabilities decrease when the avoided set grows. The induction hypothesis therefore gives the pointwise four-functions inequality indexed by $I, J, I \cap J, I \cup J$.

Let $\pi(I)$ denote the product probability of the exposed state I . The identity

$$\pi(I)\pi(J) = \pi(I \cap J)\pi(I \cup J)$$

turns that pointwise inequality into the hypothesis of the Ahlswede–Daykin four-functions theorem [1]. Summing over I and J proves (3). This completes the induction. $\square$

The proof also gives a recorded-cluster form. The functions f and g may be increasing functions of $\mathcal{K}_s$, rather than only of its support. In the inductive step, whenever the cluster avoids the exposed set Z , none of the open hyperedges incident to Z belongs to its record. Thus the original record is exactly the record in the induced model on $U \setminus Z$, and the four-functions comparison is unchanged. The same statement holds for a finite source set: adjoin a deterministic hyperedge from a new root to that set and apply the singleton statement.

The next result concerns two source records. A function of two records is called increasing-decreasing when it is increasing in the first record and decreasing in the second.

Lemma 3.2 (Two-cluster inequality). *Let $S, T \subseteq U$ be disjoint finite source sets in an independent hyperedge model. Conditional on $C_S \cap T = \emptyset$, two nonnegative increasing-decreasing functions of $(\mathcal{K}_S, \mathcal{K}_T)$ are positively associated.*

Proof. Condition first on $\mathcal{K}_S = J$, and let $K = C_S$ be its support. Every hyperedge meeting both K and its complement is then closed, while the hyperedges disjoint from K remain independent. Conditional on J , both functions decrease as the remaining coordinates enlarge $\mathcal{K}_T$. Harris' inequality gives positive association within this conditional law.

If $J \subseteq J_1$, then the corresponding support and the family of deleted hyperedges grow. Couple the two residual models with the same coordinate variables. The record $\mathcal{K}_T$ can only decrease. Hence the two conditional means increase with J : their first argument grows and their second argument shrinks. The recorded-cluster form of Lemma 3.1, applied to the law of $\mathcal{K}_S$ conditioned to avoid T , positively associates these means. Averaging the conditional Harris inequality proves the claim. This is the hyperedge version of the two-cluster conditional association argument of van den Berg, Häggström, and Kahn [3]. $\square$

4 Labelled sources

The induction needed for gluing exposes all hyperedges incident to a common source. Recording only the resulting vertex trace is insufficient because the second relation in (2) may be strict. We retain the names of the exposed hyperedges and attach their traces as anchors.

Let L be a finite label set, and let $\alpha : L \rightarrow 2^U$ assign an anchor set to each label. For $Q \subseteq L$ and $W \subseteq U$, write

$$[Q]_W = W \cap \bigcup_{\ell \in Q} \alpha(\ell).$$

Thus Q denotes the ordinary vertex source $[Q]_W$ in the hypergraph induced on W . Different labels may have equal or overlapping anchors. Retaining distinct labels when their anchors overlap is precisely what preserves intersections.

We use the following convention in induced hypergraphs. If a named vertex does not belong to W , then its cluster in W is empty and a connection to that vertex is false. Fix vertices $x, o, v \in U$, a set $A \subseteq U$, and a function $F : 2^U \rightarrow [0, \infty)$. For $Q \subseteq L$, let $R_Q^W = W \setminus C_{[Q]_W}^W$ and define

$$E_A^W(Q) = \mathbb{P}_W(o \leftrightarrow v, v \not\leftrightarrow A \cup [Q]_W), \quad (4)$$

$$M_A^W(Q) = \mathbb{P}_W(v \not\leftrightarrow A \cup [Q]_W), \quad (5)$$

$$Y_F^W(Q) = \mathbb{E}_W[F(R_Q^W); x \notin C_{[Q]_W}^W]. \quad (6)$$

The same notation with a vertex set $B \subseteq W$ in place of Q means that the source is B . Let

$$q_A(W) = \begin{cases} E_A^W(\emptyset)/M_A^W(\emptyset), & M_A^W(\emptyset) > 0, \\ 0, & M_A^W(\emptyset) = 0, \end{cases}$$

and define

$$X_F^W(Q) = \mathbb{E}_W[q_A(R_Q^W)F(R_Q^W); x \notin C_{[Q]_W}^W]. \quad (7)$$

The next lemma supplies the one-source estimate required by the labelled induction. Its proof is included because the stopping argument is unchanged when graph edges are replaced by hyperedges.

Lemma 4.1 (Stopped covariance). *Let U carry an independent hyperedge model. Let $x, v \in U$, let $S \subseteq U$ satisfy $x \in S$ and $v \notin S$, and let $f : 2^U \rightarrow [0, \infty)$ be increasing. For $W \subseteq U$, let*

$$H(W) = \text{Cov}_W(f(C_x^W), \mathbf{1}\{v \leftrightarrow S\}),$$

with $H(W) = 0$ when $v \notin W$. Then $H(W) \geq 0$, and for every $B \subseteq W$,

$$Y_H^W(B)M_S^W(\emptyset) \leq M_S^W(B)H(W). \quad (8)$$

Proof. Harris' inequality gives $H(W) \geq 0$. We prove (8) in one fixed world W ; the case $M_S^W(\emptyset) = 0$ is immediate because the left side then vanishes and the right side is nonnegative. Suppose therefore that this probability is positive. Explore C_B^W , and let $\mathcal{F}_B$ be the information read by the exploration. Put

$$G = f(C_x^W), \quad J = \mathbf{1}\{v \leftrightarrow S\}, \quad Z = \mathbf{1}\{v \nleftrightarrow S, v \leftrightarrow B\}, \quad \hat{G} = \mathbb{E}_W[G \mid \mathcal{F}_B].$$

Exact cluster conditioning and the law of total covariance give

$$H(W) = Y_H^W(B) + \text{Cov}_W(\hat{G}, J). \quad (9)$$

Indeed, on the branch $x \notin C_B^W$ the conditional covariance left after the exploration is $H(W \setminus C_B^W)$; on the complementary branch the owner's cluster has already been determined.

The event with indicator $J + Z$ is $\{v \leftrightarrow S \cup B\}$ and is increasing. The decision-tree Harris inequality [5, Theorem 3.2], applied to the tree that explores C_B^W , yields

$$\text{Cov}_W(\hat{G}, J + Z) \geq 0.$$

The variable Z is $\mathcal{F}_B$ -measurable, so $\text{Cov}_W(\hat{G}, Z) = \text{Cov}_W(G, Z)$. Let $D = \{v \nleftrightarrow S\}$, let $a = \mathbb{P}_W(D)$, and let $r = \mathbb{P}_W(Z)$. Conditional on D , the clusters of S and v are disjoint. Apply Lemma 3.2 to G and $1 - \mathbf{1}\{v \leftrightarrow B\}$. The first function increases with $\mathcal{K}_S$, whereas the second decreases with $\mathcal{K}_{\{v\}}$. The resulting positive association is equivalent to negative association between G and $\mathbf{1}\{v \leftrightarrow B\}$. Hence

$$\text{Cov}_W(G, Z) \leq \frac{r}{a} \text{Cov}_W(G, \mathbf{1}_D) = -\frac{r}{a} H(W).$$

Combining this inequality with (9) gives

$$Y_H^W(B) \leq \left(1 - \frac{r}{a}\right) H(W).$$

Since $a - r = M_S^W(B)$, multiplication by $a = M_S^W(\emptyset)$ proves (8). $\square$

The stopped estimate is the only analytic input to the next induction. The rest is an application of the four-functions theorem on the set of exposed hyperedge labels.

Lemma 4.2 (Labelled two-source comparison). *Let U carry an independent hyperedge model. Let $x, o, v \in U$, let $A \subseteq U$, and let $F : 2^U \rightarrow [0, \infty)$ satisfy $F(W) = 0$ when $v \notin W$. Suppose that, for every $W \subseteq U$ and every $B \subseteq W$,*

$$Y_F^W(B)M_A^W(\emptyset) \leq M_A^W(B)F(W). \quad (10)$$

Let L be a finite label set with anchor map $\alpha : L \rightarrow 2^U$. Then, for all $N, N' \subseteq L$,

$$E_A^U(N)Y_F^U(N') \leq M_A^U(N \cup N')X_F^U(N \cap N'). \quad (11)$$

Proof. We use strong induction on $|U|$, simultaneously for all finite label sets and anchor maps. If $v \in A \cup [N]_U$ or $o \in [N]_U$, then $E_A^U(N) = 0$. If $x \in [N']_U$, then $Y_F^U(N') = 0$. If $v \notin U$, then the assumption on F also makes $Y_F^U(N') = 0$. We may therefore remove these cases at every stage of the induction.

Suppose first that $N \cap N' = \emptyset$. Apply Lemma 3.1 to the avoided sets $A \cup [N]_U$ and $A \cup [N']_U$, using the constant function 1 and the indicator $\mathbf{1}\{o \in C_v\}$. Although the label sets are disjoint, their anchors may overlap. The intersection of the two avoided vertex sets is

$$A \cup ([N]_U \cap [N']_U),$$

which contains A . Monotonicity in the avoided set therefore gives

$$M_A^U(N')E_A^U(N) \leq E_A^U(\emptyset)M_A^U(N \cup N'). \quad (12)$$

Multiply (10), with $B = [N']_U$, by $E_A^U(N)$ and use (12). If $M_A^U(\emptyset) > 0$, division by this factor and $X_F^U(\emptyset) = q_A(U)F(U)$ prove (11). If the factor vanishes, then $E_A^U(N) = 0$, so the conclusion is again immediate.

Now let $Z = N \cap N'$ be nonempty and let $K = [Z]_U$. Labels with empty anchors may be discarded, so K is nonempty. The zero cases already removed show that $x, o, v \notin K$. Expose all hyperedges incident to K , delete K , and write E_K for the exposed label set. In $U \setminus K$ use the new label set

$$(L \setminus Z) \dot{\cup} E_K.$$

An old label ℓ receives anchor $\alpha(\ell) \setminus K$, while a fresh label $e \in E_K$ receives anchor $\iota(e) \setminus K$. The avoided set in the smaller world is $A \setminus K$; retaining A in the notation gives the same events.

For exposed states $I, J \subseteq E_K$, define

$$\hat{N}_I = (N \setminus Z) \dot{\cup} I, \quad \hat{N}'_J = (N' \setminus Z) \dot{\cup} J.$$

The two kinds of labels form a disjoint union, so

$$\hat{N}_I \cup \hat{N}'_J = ((N \cup N') \setminus Z) \dot{\cup} (I \cup J), \quad (13)$$

$$\hat{N}_I \cap \hat{N}'_J = I \cap J. \quad (14)$$

These are identities of labels, even when two fresh labels have the same anchor.

Let $\pi(I)$ be the product probability of the exposed state I . Lemma 2.2 and exact cluster conditioning give

$$\begin{aligned} E_A^U(N) &= \sum_{I \subseteq E_K} \pi(I)e(I), & Y_F^U(N') &= \sum_{J \subseteq E_K} \pi(J)y(J), \\ M_A^U(N \cup N') &= \sum_{I \subseteq E_K} \pi(I)m(I), & X_F^U(N \cap N') &= \sum_{I \subseteq E_K} \pi(I)\chi(I), \end{aligned}$$

where the four functions are the corresponding quantities in $U \setminus K$ with the sources displayed in (13) and (14). The induction hypothesis gives, for every $I, J \subseteq E_K$,

$$e(I)y(J) \leq m(I \cup J)\chi(I \cap J).$$

Moreover,

$$\pi(I)\pi(J) = \pi(I \cap J)\pi(I \cup J).$$

The Ahlswede–Daykin four-functions theorem [1] now gives (11). Since K is nonempty, the induction has passed to a strictly smaller vertex set, which completes the proof. $\square$

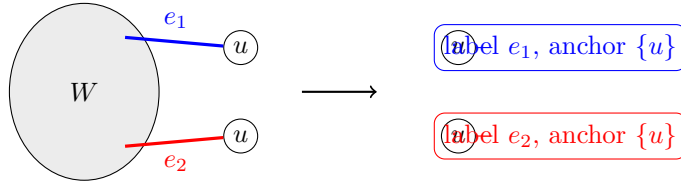


Figure 2: Two incident hyperedges can have the same outside trace. The vertex trace forgets which hyperedge was open. Separate labels preserve the intersection $I \cap J$ even when their anchors overlap.

5 Finite gluing

We now prove the finite inequality used in the lattice argument. The proof is organized around first relays. This section includes the conditioned covariance statement that controls the selection bias.

Let $F : 2^V \rightarrow \mathbb{R}$ be increasing and nonnegative. For a relay $a \in A$, write

$$m_a = \mathbb{E}F(C_a).$$

Choose a total order $\prec$ on A such that $a \prec a_1$ implies $m_a \leq m_{a_1}$. Define

$$P_a^u = \{u \leftrightarrow a\} \cap \bigcap_{a_1 \prec a} \{u \not\leftrightarrow a_1\}. \quad (15)$$

These events are disjoint and their union is $\{u \leftrightarrow A\}$.

Theorem 5.1 (First-relay inequality). *For every finite independent hyperedge model, every vertex o , every relay set $A \subseteq V$, every increasing nonnegative F , and every compatible order of the relays,*

$$\mathbb{E}[F(C_o); o \leftrightarrow A] \geq \sum_{a \in A} \mathbb{P}(P_a^o) \mathbb{E}F(C_a). \quad (16)$$

The conclusion includes hyperedge probabilities equal to zero or one.

We prove the theorem from two lemmas. Their formulation is included to make the dependence on Lemma 4.2 explicit.

For a relay set R , define its surplus at u by

$$\text{Sur}_u(R) = \mathbb{E}[F(C_u); u \leftrightarrow R] - \sum_{a \in R} \mathbb{P}(P_a^u) m_a. \quad (17)$$

Lemma 5.2 (Surplus transfer). *Let $R \subseteq V$, and let $o, v \in V$ with $v \notin R$. Then*

$$\mathbb{P}(v \not\leftrightarrow R) \text{Sur}_o(R) \geq \mathbb{P}(o \leftrightarrow v, v \not\leftrightarrow R) \text{Sur}_v(R). \quad (18)$$

Lemma 5.3 (Conditioned covariance hierarchy). *Let all hyperedge probabilities lie strictly between zero and one. Let $x, o, v, z_1, \dots, z_\ell$ be distinct vertices, and let $Y \subseteq V$ be disjoint from them. Conditional on $x \not\leftrightarrow Y$, define*

$$\kappa_f(u) = \text{Cov}(f(C_x), \mathbf{1}\{u \in C_x\} \mid x \not\leftrightarrow Y)$$

for an increasing function f . For $1 \leq j \leq \ell$, let

$$Y_j = Y \cup \{x, z_1, \dots, z_{j-1}\}, \quad c_j(u) = \mathbb{P}(u \in C_{z_j} \mid z_j \not\leftrightarrow Y_j).$$

Define $L^0 h = h$ and

$$(L^j h)(u) = (L^{j-1} h)(u) - c_j(u)(L^{j-1} h)(z_j). \quad (19)$$

Finally, let

$$\tau = \mathbb{P}(o \in C_v \mid v \not\leftrightarrow Y \cup \{x, z_1, \dots, z_\ell\}).$$

Then

$$(L^\ell \kappa_f)(o) \geq \tau(L^\ell \kappa_f)(v). \quad (20)$$

Proof. We may add a constant to f , so we assume that f is nonnegative. We prove the hierarchy by strong induction on ℓ . The proof has three components: passing between a conditioned covariance and its residual worlds, expanding the level form, and comparing the two observer coefficients.

Step 1: residual worlds. Let $\nu = \mathbb{P}(\cdot \mid x \not\leftrightarrow Y)$, and under ν expose C_Y . The remaining vertex set

$$W = V \setminus C_Y$$

contains x . Conditional on W , the hyperedges contained in W form the original independent model induced on W . Write π for the law of W under ν , and put

$$\kappa_f^W(u) = \text{Cov}_W(f(C_x^W), \mathbf{1}\{u \in C_x^W\}).$$

The constants c_j and τ in the level form remain the global constants from the statement. Define the residual-world margin by

$$\mathcal{W}_\ell(f) = \mathbb{E}_\pi \left[(L^\ell \kappa_f^W)(o) - \tau(L^\ell \kappa_f^W)(v) \right]. \quad (21)$$

We record why nonnegativity of (21) implies the desired inequality. The level operators and the final observer subtraction determine constants λ_u , indexed by $u \in \{o, v, z_1, \dots, z_\ell\}$, such that

$$\mathfrak{M}_\ell(f) = \text{Cov}_\nu \left(f(C_x), \sum_u \lambda_u \mathbf{1}\{u \in C_x\} \right).$$

Under ν , write $A = C_x$ and $B = C_Y$. Let P_A and P_B denote conditional expectation given A and B , respectively. The two-block Gibbs operator on functions of A is

$$\mathcal{A} = P_A P_B, \quad (\mathcal{A}f)(A) = \mathbb{E}_\nu[\mathbb{E}_\nu[f(A) \mid B] \mid A].$$

The test function in the covariance is measurable with respect to A . Conditioning first on B therefore gives the exact identity

$$\mathfrak{M}_\ell(f) = \mathcal{W}_\ell(f) + \mathfrak{M}_\ell(\mathcal{A}f), \quad (22)$$

because the difference of the two covariances is

$$\mathbb{E}_\nu \left[\text{Cov}_\nu \left(f(A), \sum_u \lambda_u \mathbf{1}\{u \in A\} \mid B \right) \right] = \mathcal{W}_\ell(f).$$

The coupling from the proof of Lemma 3.2 shows that $B \mapsto \mathbb{E}_\nu[f(A) \mid B]$ is decreasing when f is increasing. During the next half-step, a larger A deletes more hyperedges and produces a smaller B . Thus $\mathcal{A}$ preserves increasing functions. It also preserves nonnegativity and constants. Because each hyperedge probability is less than one, the B -resampling closes every hyperedge incident to Y with probability at least

$$\epsilon_Y = \prod_{e: \iota(e) \cap Y \neq \emptyset} (1 - p_e) > 0.$$

This event forces $B = Y$. The following A -resampling then has a fixed law, independent of the old state. Consequently,

$$\text{osc}(\mathcal{A}^n f) \leq (1 - \epsilon_Y)^n \text{osc}(f).$$

When Y is empty, the first application of $\mathcal{A}$ is already constant. In both cases $\mathcal{A}^n f$ converges uniformly to a constant. Iterating (22) now gives

$$\mathfrak{M}_\ell(f) = \sum_{i=0}^{n-1} \mathcal{W}_\ell(\mathcal{A}^i f) + \mathfrak{M}_\ell(\mathcal{A}^n f).$$

The last term tends to zero. Hence it is enough to prove $\mathcal{W}_\ell(g) \geq 0$ for every increasing nonnegative g .

Step 2: exact expansion. Fix such a function g and let

$$X = \{x, z_1, \dots, z_\ell\}.$$

For $r \in \{o, v\}$ define

$$h_r(W) = \text{Cov}_W(g(C_x^W), \mathbf{1}\{r \leftrightarrow X\}).$$

Let C_x^{-K} denote the cluster of x after all hyperedges meeting a vertex set K have been deleted, and define C_Y^{-K} in the same way. We will use the function

$$\Phi(K) = \mathbb{E} \left[\mathbf{1}\{x \not\leftrightarrow Y \text{ after deleting } K\} \left(g(C_x^{-C_Y^{-K}}) - g(C_x^{-K}) \right) \right]. \quad (23)$$

In the first cluster in (23), we delete the hyperedges meeting C_Y^{-K} from the original configuration; hyperedges meeting K are therefore available unless they also meet that cluster.

For $K \subseteq K_1$, deleting K_1 shrinks $C_Y^{-K_1}$ and $C_x^{-K_1}$. The first cluster in (23) can only grow, while the second can only shrink. The avoidance event can only grow as well. Thus the integrand is pointwise nonnegative and increasing in K . It follows that Φ is an increasing nonnegative function.

For $1 \leq j \leq \ell$, let ρ_j be the law conditional on $z_j \not\leftrightarrow Y_j$, and let $L_{>j}$ be the level form made from $z_{j+1}, \dots, z_\ell$ with the same constants $c_{j+1}, \dots, c_\ell$ and τ . Repeated use of the disjoint identity

$$\{u \leftrightarrow S \cup \{z\}\} = \{u \leftrightarrow S\} \dot{\cup} \{u \not\leftrightarrow S, u \leftrightarrow z\} \quad (24)$$

and exact conditioning on C_{z_j} gives

$$\mathcal{W}_\ell(g) = \mathcal{H}_\ell(g) + \sum_{j=1}^{\ell} \frac{\mathbb{P}(z_j \not\leftrightarrow Y_j)}{\mathbb{P}(x \not\leftrightarrow Y)} \mathfrak{M}_{>j}^{(z_j, Y_j)}(\Phi), \quad (25)$$

where

$$\mathcal{H}_\ell(g) = \mathbb{E}_\pi[h_o(W) - \tau h_v(W)]$$

and

$$\mathfrak{M}_{>j}^{(z_j, Y_j)}(\Phi) = (L_{>j} \kappa_{\Phi, j})(o) - \tau (L_{>j} \kappa_{\Phi, j})(v), \quad \kappa_{\Phi, j}(u) = \text{Cov}_{\rho_j}(\Phi(C_{z_j}), \mathbf{1}\{u \in C_{z_j}\}).$$

For completeness, we indicate every substitution behind (25). Apply (24) successively to $S = \{x, z_1, \dots, z_{j-1}\}$. At stage j , subtracting $c_j(u)$ centers the indicator of $u \in C_{z_j}$ under ρ_j . Conditional on $C_{z_j} = K$, exact cluster conditioning identifies the conditional mean of the centered residual with $-\Phi(K)$. The minus sign cancels the minus from subtracting the decoy term. Changing from the conditional covariance under ρ_j to the original measure contributes $\mathbb{P}(z_j \not\leftrightarrow Y_j)$, while the definition of π contributes $\mathbb{P}(x \not\leftrightarrow Y)^{-1}$. The later substitutions are exactly the operator $L_{>j}$. This proves (25) as an identity, with no inequality used in the expansion.

Each margin in the sum in (25) has $\ell - j < \ell$ decoys. Its owner is z_j , its avoided set is Y_j , and its final avoided set is

$$Y_j \cup \{z_j, z_{j+1}, \dots, z_\ell\} = Y \cup X.$$

Its observer coefficient is therefore the same τ . The induction hypothesis and the monotonicity of Φ show that every summand is nonnegative.

Step 3: the horizontal term. It remains to prove $\mathcal{H}_\ell(g) \geq 0$. In a fixed world W , let

$$q(W) = \mathbb{P}_W(o \leftrightarrow v \mid v \not\leftrightarrow X).$$

We first show

$$h_o(W) \geq q(W) h_v(W). \quad (26)$$

Write $m = \mathbb{E}_W g(C_x^W)$, let $D = \{v \not\leftrightarrow X\}$, and let $J = \{o \leftrightarrow v, v \not\leftrightarrow X\}$. Since

$$\{o \leftrightarrow X \cup \{v\}\} = \{o \leftrightarrow X\} \dot{\cup} J,$$

Harris' inequality gives

$$h_o(W) \geq -\mathbb{E}_W[(g(C_x^W) - m); J].$$

Conditional on D , apply Lemma 3.2 to $g(C_x)$ and $1 - \mathbf{1}\{o \in C_v\}$. The first function increases with the record $\mathcal{K}_X$, and the second decreases with $\mathcal{K}_{\{v\}}$. Equivalently, $g(C_x)$ and $\mathbf{1}\{o \in C_v\}$ are negatively associated. Hence

$$\mathbb{P}_W(D) \mathbb{E}_W[(g(C_x^W) - m); J] \leq \mathbb{P}_W(J) \mathbb{E}_W[(g(C_x^W) - m); D].$$

Now $h_v(W) = -\mathbb{E}_W[(g(C_x^W) - m); D]$ and $q(W) = \mathbb{P}_W(J)/\mathbb{P}_W(D)$, proving (26).

The world price $q(W)$ still has to be replaced by the global price τ . Apply Lemma 4.2 with $A = X$ and $F(W) = h_v(W)$. Give each vertex of Y its own singleton label, and take the same label set for N and N' . Lemma 4.1 supplies (10); Harris gives $F \geq 0$, and $F(W) = 0$ when $v \notin W$. The diagonal conclusion is

$$\mathbb{E}[q(W) h_v(W); x \not\leftrightarrow Y] \geq \tau \mathbb{E}[h_v(W); x \not\leftrightarrow Y]. \quad (27)$$

After division by $\mathbb{P}(x \not\leftrightarrow Y)$, equations (26) and (27) give $\mathcal{H}_\ell(g) \geq 0$.

We have proved that the horizontal term and every lower-level term in (25) are nonnegative. Thus $\mathcal{W}_\ell(g) \geq 0$. Step 1 converts this residual-world statement to $\mathfrak{M}_\ell(f) \geq 0$, completing the strong induction and proving (20). $\square$

Proof of Lemma 5.2. First suppose that all probabilities lie strictly between zero and one. We prove a stronger assertion that retains a list of decoys. For a relay set R and a list $Z = (z_1, \dots, z_\ell)$ outside R , build the level operator as in (19), but condition z_j to avoid

$$R \cup \{z_1, \dots, z_{j-1}\}.$$

Use the observer coefficient

$$\mathbb{P}(o \leftrightarrow v \mid v \not\leftrightarrow R \cup \{z_1, \dots, z_\ell\}).$$

Let $\mathcal{S}_{R,Z}(h)$ denote the resulting level form at o minus this coefficient times the level form at v . We claim

$$\mathcal{S}_{R,Z}(u \mapsto \text{Sur}_u(R)) \geq 0 \tag{28}$$

for every decoy list satisfying the distinctness conditions of Lemma 5.3. We prove the claim by induction on $|R|$, simultaneously for all such lists. The empty relay set gives equality.

For the induction step, let k be the last relay and let $T = R \setminus \{k\}$. Write

$$D_k = \{k \not\leftrightarrow T\}, \quad c_k(u) = \mathbb{P}(u \leftrightarrow k \mid D_k),$$

and define

$$\gamma_k = \text{Cov}(F(C_k), \mathbf{1}\{k \leftrightarrow T\}).$$

Splitting according to the first relay gives the exact identity

$$\text{Sur}_u(R) = \text{Sur}_u(T) + \mathbb{P}(D_k) \text{Cov}(F(C_k), \mathbf{1}\{u \in C_k\} \mid D_k) - \gamma_k c_k(u). \tag{29}$$

Apply $\mathcal{S}_{R,Z}$ to (29). Lemma 5.3 with owner k , avoided set T , decoy list Z , and the cluster functional induced by F shows that the margin of the conditional-covariance term is nonnegative. The same lemma, applied to $\mathbf{1}\{C_k \neq \{k\}\}$, shows that

$$\mathcal{S}_{R,Z}(c_k) \geq 0. \tag{30}$$

Indeed, its conditional covariance profile is a positive constant times c_k ; the constant is the conditional probability that every hyperedge incident to k is closed.

Compatibility of the relay order gives

$$\gamma_k \leq \text{Sur}_k(T), \tag{31}$$

because the difference is

$$\sum_{a \in T} \mathbb{P}(P_a^k)(m_k - m_a) \geq 0.$$

The definitions of the level operators give the algebraic identity

$$\mathcal{S}_{R,Z}(\text{Sur}(T)) - \text{Sur}_k(T) \mathcal{S}_{R,Z}(c_k) = \mathcal{S}_{T,(k,Z)}(\text{Sur}(T)). \tag{32}$$

The right side is nonnegative by the induction hypothesis. Equations (29)–(32) therefore give

$$\mathcal{S}_{R,Z}(\text{Sur}(R)) \geq \mathcal{S}_{T,(k,Z)}(\text{Sur}(T)) \geq 0.$$

This proves (28).

Take Z to be empty. If o and v are distinct and lie outside R , then multiplying (28) by $\mathbb{P}(v \not\leftrightarrow R) > 0$ gives (18). If $o = v$, the two sides are equal. If $o \in R$, the event on the right of (18) is empty, and $\text{Sur}_o(R) \geq 0$: in each first-relay event, the selected relay a has $m_a \leq m_o$, while $C_a = C_o$. Thus the transfer inequality holds for all positions allowed in its statement when the probabilities lie in the open cube.

For probabilities on the boundary of $[0, 1]^E$, use the rank-free formula

$$\text{Sur}_u(R) = \mathbb{E} \left[\left(F(C_u) - \min_{a \in C_u \cap R} m_a \right) \mathbf{1}\{u \leftrightarrow R\} \right]. \quad (33)$$

Approximate the weight vector by vectors in $(0, 1)^E$. There are finitely many relay orders, so a subsequence has a fixed compatible order. Every term in the denominator-free form of (18) is a polynomial in the hyperedge probabilities. Continuity proves the boundary cases. $\square$

Proof of Theorem 5.1. We prove by induction on $|A|$ that $\text{Sur}_o(A) \geq 0$. The empty case is immediate. Let k be the last relay, and let $T = A \setminus \{k\}$. Put

$$D = \{k \not\leftrightarrow T\}, \quad H = \{o \leftrightarrow k, k \not\leftrightarrow T\}.$$

Splitting off the first-relay event of k gives

$$\text{Sur}_o(A) = \text{Sur}_o(T) - \mathbb{E}[(m_k - F(C_k)); H]. \quad (34)$$

Conditional positive association from Lemma 3.1 yields

$$\mathbb{P}(D)\mathbb{E}[(m_k - F(C_k)); H] \leq \mathbb{P}(H)\gamma, \quad (35)$$

where

$$\gamma = m_k \mathbb{P}(D) - \mathbb{E}[F(C_k); D].$$

The first-relay partition and compatibility give $\gamma \leq \text{Sur}_k(T)$. Lemma 5.2 gives

$$\mathbb{P}(D) \text{Sur}_o(T) \geq \mathbb{P}(H) \text{Sur}_k(T).$$

Combining this inequality with (34)–(35) proves $\mathbb{P}(D) \text{Sur}_o(A) \geq 0$. If $\mathbb{P}(D) > 0$, the result follows by division. If $\mathbb{P}(D) = 0$, then $\mathbb{P}(H) = 0$ and the induction hypothesis gives the result. This completes the induction and proves (16). $\square$

Theorem 5.4 (Hyperedge gluing). *Let $o, b \in V$, and let $A \subseteq V$ be nonempty. In every finite independent hyperedge model,*

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \min_{a \in A} \mathbb{P}(a \leftrightarrow b). \quad (36)$$

Proof. Apply Theorem 5.1 to $F(K) = \mathbf{1}\{b \in K\}$. The first-relay events partition $\{o \leftrightarrow A\}$, so

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \geq \sum_{a \in A} \mathbb{P}(P_a^o) \mathbb{P}(a \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \min_{a \in A} \mathbb{P}(a \leftrightarrow b).$$

Dropping $\{o \leftrightarrow A\}$ from the event on the left proves (36). $\square$

6 Pinned site gluing

We now return to ordinary site percolation on a finite graph. Vertices may have unequal opening probabilities.

Corollary 6.1 (Pinned site gluing). *Let G be a finite graph with independent site probabilities. Let $o, b \in V(G)$, and let $A \subseteq V(G)$ be nonempty. Suppose that o , b , and every vertex of A are open with probability one. Then*

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \min_{a \in A} \mathbb{P}(a \leftrightarrow b). \quad (37)$$

Consequently, if both factors on the right exceed $1 - \delta$, then $\mathbb{P}(o \leftrightarrow b) > (1 - \delta)^2$.

Proof. Use the hyperedge representation in Lemma 2.1. All connectivity events in (37) agree with their hyperedge counterparts, including possible coincident terminals because the named vertices are forced open. Theorem 5.4 proves the first claim. The second follows by multiplication. $\square$

Two wiring operations will be used below. To wire a source set S , add a new vertex o that is open with probability one and adjacent to each vertex of S . Then $o \leftrightarrow T$ is exactly the event that S connects to T . To wire a target set T , add a new vertex b that is open with probability one and adjacent to each vertex of T . Then $S \leftrightarrow b$ is exactly the event that S connects to T . Wiring T can increase the probability that the source reaches the relay set; this only strengthens the hypothesis in Corollary 6.1.

Conditioning on finitely many site states gives another operation. Delete the closed conditioned vertices and contract each connected component of open conditioned vertices to one vertex that is open with probability one. The unconditioned vertices remain independent with their original probabilities. Connections among the remaining vertices are unchanged. Thus the source, target, and relays in the conditional graph satisfy the pinning hypothesis of Corollary 6.1.

7 The site reduction

We now prove the infinite-volume implication used at criticality. The argument adapts the same-parameter block construction of Kozma and Nitzan [7, Sections 2–4] to site variables. We include the finite statements, the site-specific separation of the revealed variables, and the parameter order because the conclusion must hold at the same value of p .

For $m \geq 0$, put $\Lambda_m = [-m, m]^d \cap \mathbb{Z}^d$. If $B = [a, b] \cap \mathbb{Z}^d$ is an integer box, write

$$B^{+r} = [a - r\mathbf{1}, b + r\mathbf{1}] \cap \mathbb{Z}^d.$$

For a finite set $K \subseteq \mathbb{Z}^d$, its inner and outer vertex boundaries are

$$\begin{aligned} \partial_{\text{in}} K &= \{x \in K : \|x - y\|_1 = 1 \text{ for some } y \notin K\}, \\ \partial_{\text{out}} K &= \{y \notin K : \|x - y\|_1 = 1 \text{ for some } x \in K\}. \end{aligned}$$

For sets $S, T, D \subseteq \mathbb{Z}^d$, the notation $S \overset{D}{\leftrightarrow} T$ means that an open site path contained in D has one endpoint in each of S and T .

7.1 Finite geometry

A **site subbox at parameter** p consists of a finite graph G , an integer box $D \subseteq V(G) \cap \mathbb{Z}^d$, and independent site states such that:

- (S1) the sites of D have opening probability p ;
- (S2) the graph induced by D is the nearest-neighbour lattice graph;
- (S3) an edge of G from $V(G) \setminus D$ to D ends at the inner vertex boundary of D .

The sites outside D may have different opening probabilities. Wiring and conditioning are allowed there. A source o in a site subbox is always outside D and is forced open.

A **geometry** $\gamma = (Q, F)$ is a pair of integer boxes. At least one coordinate of F lies entirely in $[1, \infty)$ or entirely in $(-\infty, -1]$. Define

$$Q_\gamma(\ell, v) = v + \ell Q, \quad F_\gamma(\ell, v) = v + \ell F.$$

The geometry is **p -hittable** when, for each $\alpha > 0$, there are integers $k, \ell_0 \geq 1$ such that, for all integers $m \geq k$ and $\ell \geq \ell_0$,

$$\mathbb{P}_p(\Lambda_m \overset{Q_\gamma(\ell, 0)}{\leftrightarrow} F_\gamma(\ell, 0)) > 1 - \alpha. \quad (38)$$

Let $B \subseteq D$ be an integer box, let $R \geq 1$ be an integer, and let $\mathcal{H}$ be a finite list of geometries. A nonempty set $T \subseteq D$ is a **target for** $(B, D, R, \mathcal{H})$ if, for each $v \in B^{+R}$, there are $\ell \geq R$ and $\gamma \in \mathcal{H}$ such that

$$Q_\gamma(\ell, v) \subseteq D, \quad F_\gamma(\ell, v) \subseteq T. \quad (39)$$

These quantifiers are uniform in v .

We use two standard finite-size facts for supercritical site percolation. The first follows from uniqueness, and the second is the square-root argument for the orthants of a face; compare [6, Sections 7.2 and 8.1]. We state and prove the precise forms used below.

Lemma 7.1 (Local site facts). *Let $d \geq 2$ be an integer, and let $p \in (0, 1)$ satisfy $\theta_{\text{site}}(p) > 0$.*

(L1) *For $m \geq 1$ and $\alpha > 0$, there is $M > m$ such that, with probability greater than $1 - \alpha$, all open components in $v + \Lambda_M$ that meet both $v + \Lambda_m$ and the inner boundary of $v + \Lambda_M$ belong to one component of $v + \Lambda_M$. The estimate is uniform in v .*

(L2) *Let $i \in \{1, \dots, d\}$ and $\epsilon \in \{-1, 1\}^d$. In the cube $Q = [-1, 1]^d$, set*

$$F_{i, \epsilon} = \{z \in Q : \epsilon_i z_i = 1, \ 0 \leq \epsilon_j z_j \leq 1 \ (j \neq i)\}.$$

Each pair $(Q, F_{i, \epsilon})$ is p -hittable.

Proof. The infinite open cluster is almost surely unique [4]. Fix two vertices $a, b \in v + \Lambda_m$. If one of their full-lattice clusters is finite, then that cluster misses the boundary of $v + \Lambda_M$ for all sufficiently large M . If both clusters are infinite, then uniqueness joins a to b by a finite path, which lies in $v + \Lambda_M$ for all sufficiently large M . Thus the probability that a and b lie in two distinct components of $v + \Lambda_M$, both reaching its inner boundary, tends to zero. A union bound over the finitely many pairs $a, b \in v + \Lambda_m$ proves (L1). Translation invariance makes the choice of M uniform in v .

We prove (L2) by the square-root argument. Since $\theta_{\text{site}}(p) > 0$, translation ergodicity and uniqueness imply that the unique infinite cluster exists almost surely. The increasing boxes

Λ_m therefore meet it with probability tending to one. Let $q = d2^d$ be the number of indexed orthant-faces in the statement. Choose m_0 so that

$$\mathbb{P}_p(\Lambda_{m_0} \not\leftrightarrow \infty) < \alpha^q.$$

For every $\ell \geq m_0$, an infinite path starting in Λ_{m_0} contains a path in Λ_ℓ from Λ_{m_0} to $\partial_{\text{in}}\Lambda_\ell$. Hence the probability that no such path reaches the full inner boundary is less than α^q .

For each indexed orthant-face, let $D_{i,\epsilon}$ be the decreasing event that it is not reached from Λ_{m_0} inside Λ_ℓ . The intersection of the q events $D_{i,\epsilon}$ is precisely the event that the full inner boundary is not reached. Harris' inequality for decreasing site events and lattice symmetry give

$$\mathbb{P}_p(D_{i,\epsilon})^q \leq \mathbb{P}_p\left(\bigcap_{i,\epsilon} D_{i,\epsilon}\right) < \alpha^q.$$

Thus every orthant-face is reached with probability greater than $1 - \alpha$. Enlarging the source box only enlarges the event. We may take $k = \ell_0 = m_0$ in (38). $\square$

The following elementary lattice construction separates the site variables used for a seed from those later fixed by conditioning. This separation is the site-percolation change that is absent from the bond proof.

Lemma 7.2 (Shell windows). *Let $d \geq 2$, $M \geq 1$, and $j \geq 2M + 2$ be integers. Let B be a nonempty integer box in $\mathbb{Z}^d$, and suppose every side of B^{+j} has length at least $2M + 3$. Put*

$$S_j = B^{+(j-1)} \setminus B^{+(j-2M-2)}.$$

For each $x \in \partial_{\text{out}}B^{+j}$ there are a point v_x , a face U_x , and a finite seed set J_x with the following properties:

- (W1) $v_x + \Lambda_M \subseteq S_j$, and U_x is the outward face of this cube in the direction of x ;
- (W2) $J_x \subseteq B^{+j} \setminus B^{+(j-1)}$ and $|J_x| \leq s_d(M) := (2M + 5)^d$;
- (W3) if x and all sites of J_x are open, then x is connected to each open site of U_x using $\{x\} \cup J_x \cup U_x$;
- (W4) if $\|x - y\|_\infty > 4M + 8$, then $J_x \cap J_y = \emptyset$.

Proof. Choose a coordinate i and sign σ such that $x = y + \sigma e_i$, where y lies on the corresponding face of B^{+j} . Inside that face choose a $(2M + 1)^{d-1}$ plaquette P_x containing y ; when y is near a corner, slide the centre of the plaquette toward the interior of the face. Join y to P_x inside the smallest coordinate box in that face, and put all sites of this box into J_x . Set

$$U_x = P_x - \sigma e_i, \quad v_x = \text{centre}(P_x) - \sigma(M + 1)e_i.$$

The first three claims now follow coordinate by coordinate. Notice that P_x is outside S_j , whereas U_x is its first layer inside S_j . Every site of J_x is at sup-distance at most $2M + 4$ from x , which proves (W4). The displayed cardinality bound follows from containment in a box of side $2M + 5$. $\square$

7.2 The target lemma

Lemma 7.3 (Target extension). *Let $d \geq 3$ be an integer, and let $p \in (0, 1)$ satisfy $\theta_{\text{site}}(p) > 0$. For each $\varepsilon > 0$, there is $\delta > 0$ with the following property. For every finite list $\mathcal{H}$ of p -hittable geometries, there is an integer $R \geq 1$ such that the next implication holds. Let (G, D) be a finite site subbox at parameter p , and let $o \in V(G) \setminus D$ be forced open. Let $B \subseteq D$ be a nonempty integer box satisfying $B^{+R} \subseteq D$, and let $T \subseteq D$ be a target for $(B, D, R, \mathcal{H})$. Then*

$$\mathbb{P}(o \leftrightarrow B) > 1 - \delta \implies \mathbb{P}(o \leftrightarrow T) > 1 - \varepsilon. \quad (40)$$

The number δ depends only on d, p, ε ; in particular, it is chosen before $\mathcal{H}$.

Proof. It is enough to treat $0 < \varepsilon \leq 1$. Put

$$\delta_c = \frac{\varepsilon}{8}, \quad \delta = \frac{\varepsilon \delta_c}{12}. \quad (41)$$

Then $\delta \leq \delta_c$, $3\delta \leq 1$, and

$$(1 - \delta_c)^2 > 1 - \frac{\varepsilon}{2}, \quad 12\delta = \varepsilon \delta_c. \quad (42)$$

Choose common hittability scales for the finite list $\mathcal{H}$ and for the quarter-face list in Lemma 7.1, with error δ^2 . Choose m beyond all source thresholds. Then choose $M > m$ beyond all quarter-face scale thresholds so that (L1) also has error δ^2 . Let $s = s_d(M)$. Choose k and then N so that

$$(1 - p^s)^k \leq \delta, \quad N \geq k|\Lambda_{4M+8}|. \quad (43)$$

Finally choose an integer $L \geq 1$ satisfying

$$L \geq \frac{1}{\delta(1-p)^{2dN}}. \quad (44)$$

Use the consecutive levels

$$j_t = 2M + 2 + t, \quad 0 \leq t < L.$$

Choose $R \geq 2M + L + 2$ larger than every scale threshold belonging to $\mathcal{H}$. Then $j_t + 1 \leq R$, so each contact, seed, cube, and shell used at level j_t lies in $B^{+R} \subseteq D$. This is the only point at which R depends on $\mathcal{H}$.

For a level $j = j_t$, explore the open cluster of o in the complement of B^{+j} . Let K_j be the set of reached sites in $\partial_{\text{out}} B^{+j}$, and call them contacts. If $|K_j| < N$, at most $2dN$ sites in B^{+j} are adjacent to K_j . Those sites have not been read by the exterior exploration. Closing all of them has conditional probability at least

$$q_N = (1 - p)^{2dN}$$

and blocks entry into B^{+j} .

Here is the counting argument behind the choice of j . Let X be the number of failure levels, $|K_{j_t}| < N$, among the L selected levels. On $\{o \leftrightarrow B, X \geq t\}$, expose the t outermost failure levels in order. At each level, reaching B requires at least one of the fresh inward sites to be open. Consequently

$$\mathbb{P}(o \leftrightarrow B, X \geq t) \leq (1 - q_N)^t.$$

The tail-sum formula gives

$$\sum_{t=0}^{L-1} \mathbb{P}(o \leftrightarrow B, |K_{j_t}| < N) = \mathbb{E}[X; o \leftrightarrow B] \leq \sum_{t \geq 1} (1 - q_N)^t < q_N^{-1}.$$

By (44), one level j therefore satisfies

$$\mathbb{P}(o \leftrightarrow B, |K_j| < N) < \delta.$$

Since $\mathbb{P}(o \leftrightarrow B) > 1 - \delta$, this level satisfies

$$\mathbb{P}(|K_j| \geq N) > 1 - 2\delta. \quad (45)$$

On $\{|K_j| \geq N\}$, the greedy packing bound and (43) select k contacts separated by more than $4M + 8$. Apply Lemma 7.2. Their seed sets are disjoint, are disjoint from S_j , and have not been read by the exterior exploration. Thus

$$\mathbb{P}(\text{none of the selected seed sets is open} \mid K_j) \leq (1 - p^s)^k \leq \delta. \quad (46)$$

Let $\mathcal{G}$ be the event that at least one selected contact has an open seed, and use a fixed enumeration to let F_x denote the event that x is the first such contact. The events F_x are disjoint, depend only on sites outside S_j , and satisfy

$$\sum_x \mathbb{P}(F_x) = \mathbb{P}(\mathcal{G}) > 1 - 3\delta. \quad (47)$$

Fix a potential contact x . The target property applied at v_x supplies $\ell \geq R$ and $\gamma \in \mathcal{H}$ for which (39) holds. Consider the intersection of these three events:

(G1) local uniqueness between $v_x + \Lambda_m$ and $v_x + \Lambda_M$;

(G2) $v_x + \Lambda_m \xleftrightarrow{v_x + \Lambda_M} U_x$;

(G3) $v_x + \Lambda_m \xleftrightarrow{Q_\gamma(\ell, v_x)} F_\gamma(\ell, v_x)$.

Each failure has probability below δ^2 . Hence their intersection, denoted G_x , has probability greater than $1 - 3\delta^2$.

Condition on the site states in S_j . If $\mathbb{P}(G_x \mid \omega|_{S_j}) > 1 - \delta$, choose one realization of G_x with that shell state. Its path in (G2) ends at some $u \in U_x$. This path and the state of u are fixed by the shell state. In every other realization of G_x with the same shell state, the path in (G3) leaves $v_x + \Lambda_m$; (G1) joins its last inner component to the fixed path from (G2). It follows that

$$\mathbb{P}(u \xleftrightarrow{D} T \mid \omega|_{S_j}) > 1 - \delta. \quad (48)$$

Markov's inequality applied to $1 - \mathbb{P}(G_x \mid \omega|_{S_j})$ shows that shell states for which this conclusion fails have probability below 3δ . Let H_x be the event that an open $u \in U_x$ satisfying (48) exists. We have

$$\mathbb{P}(H_x) \geq 1 - 3\delta. \quad (49)$$

It remains to average without choosing a relay in advance. For a shell pattern ξ , let

$$A_\xi = \left\{ u \in S_j : \xi(u) = 1, \mathbb{P}^\xi(u \xleftrightarrow{D} T) > 1 - \delta \right\}, \quad \phi_\xi = \mathbb{P}^\xi(o \leftrightarrow A_\xi),$$

where $\mathbb{P}^\xi$ is the product site law with the shell fixed to ξ . Let $p_\xi = \mathbb{P}(\omega|_{S_j} = \xi)$. Lemma 7.2 shows that $F_x \cap H_x$ implies $o \leftrightarrow A_\xi$. Since F_x is determined outside S_j and H_x is determined by S_j , independence, (47), and (49) give the exact averaged estimate

$$\sum_\xi p_\xi \phi_\xi \geq \sum_x \mathbb{P}(F_x \cap H_x) = \sum_x \mathbb{P}(F_x) \mathbb{P}(H_x) > (1 - 3\delta)^2. \quad (50)$$

Call a pattern bad if $\phi_\xi \leq 1 - \delta_c$. Since $1 - (1 - 3\delta)^2 \leq 6\delta$, equation (50) gives

$$\delta_c \sum_{\xi \text{ bad}} p_\xi \leq \sum_{\xi} p_\xi (1 - \phi_\xi) < 6\delta \leq \frac{\varepsilon \delta_c}{2}.$$

The bad patterns therefore have total mass below $\varepsilon/2$.

For a good pattern, wire T to a forced-open target b . Every member of A_ξ is fixed open, and o is fixed open. Moreover, $\delta \leq \delta_c$ gives

$$\mathbb{P}^\xi(o \leftrightarrow A_\xi) > 1 - \delta_c, \quad \min_{u \in A_\xi} \mathbb{P}^\xi(u \leftrightarrow b) > 1 - \delta_c.$$

Corollary 6.1 and (42) imply $\mathbb{P}^\xi(o \leftrightarrow T) > 1 - \varepsilon/2$. Averaging over good patterns gives

$$\mathbb{P}(o \leftrightarrow T) > (1 - \varepsilon/2)^2 > 1 - \varepsilon,$$

which proves (40). □

7.3 Long boxes

For a coordinate i , a sign $\sigma \in \{-1, 1\}$, and an integer $K \geq 2$, let $\gamma_{i,\sigma,K}$ be the geometry

$$\begin{aligned} Q &= \{x : -1 \leq \sigma x_i \leq K, |x_j| \leq 1 \ (j \neq i)\}, \\ F &= \{x : \sigma x_i = K, |x_j| \leq 1 \ (j \neq i)\}. \end{aligned}$$

Lemma 7.4 (Long-box hitting). *Let $d \geq 3$ be an integer, and let $p \in (0, 1)$ satisfy $\theta_{\text{site}}(p) > 0$. For every coordinate $1 \leq i \leq d$, every sign $\sigma \in \{-1, 1\}$, and every integer $K \geq 2$, the geometry $\gamma_{i,\sigma,K}$ is p -hittable.*

Proof. We include the face chain because this is the first use of the uniformity of δ in Lemma 7.3. Fix a desired error α , and let $N = 8K - 4$. Put $\eta_N = \alpha$. Recursively, for $0 \leq t < N$, choose η_t to be the input tolerance supplied by the target lemma with output tolerance η_{t+1} . Apply the lemma to the finite quarter-face list at each of these N tolerances. Let $R_\star$ be the maximum of the resulting inflation radii, and put $\beta = \eta_0$. A target for radius $R_\star$ is also a target for each smaller radius, so these choices permit all N applications below. Choose k, n_1 so that a quarter of the boundary of Λ_L is reached from Λ_k with probability greater than $1 - \beta$ whenever $L \geq n_1$; this is Lemma 7.1(L2).

For a scale r , put

$$s = \lfloor r/8 \rfloor, \quad L_0 = Kr - Ns, \quad L_t = L_0 + ts, \quad w_t = L_0 + tR_\star.$$

Take r so large that

$$s \geq 3K + 3KR_\star + k + n_1 + 8. \tag{51}$$

For $0 \leq t \leq N$, let

$$B_t = \{x : \sigma x_i = L_t, |x_j| \leq w_t \ (j \neq i)\}.$$

The initial quarter-face estimate gives a connection from Λ_k to B_0 . For $t < N$, use the site subbox

$$D_t = \{x : L_t - 2s \leq \sigma x_i \leq L_t + s, |x_j| \leq r \ (j \neq i)\}.$$

The wired source Λ_k is outside D_t . For $v \in B_t^{+R_\star}$ take

$$\ell(v) = L_t + s - \sigma v_i$$

and choose the transverse signs of a quarter-face to point toward the coordinate axes. The inequalities in (51) give

$$\ell(v) \geq R_\star, \quad Q_\gamma(\ell(v), v) \subseteq D_t, \quad F_\gamma(\ell(v), v) \subseteq B_{t+1}.$$

Here are the bounds that verify the inclusions. Since $|\sigma v_i - L_t| \leq R_\star$, we have

$$s - R_\star \leq \ell(v) \leq s + R_\star.$$

The choice of s gives $s \geq 2R_\star$. Hence the longitudinal interval of $Q_\gamma(\ell(v), v)$ lies between $L_t - 2s$ and $L_t + s$. Transversely, (51) gives

$$w_N + s + 2R_\star \leq r.$$

Thus the full transverse cube lies in D_t . On the far face, choose each transverse orthant to point toward the corresponding coordinate hyperplane. Its coordinates have absolute value at most $\max\{w_t + R_\star, s + R_\star\} \leq w_{t+1}$, proving the final inclusion. Thus B_{t+1} is a target for $(B_t, D_t, R_\star, \mathcal{Q}_d)$. Induction on t gives

$$\mathbb{P}_p \left(\Lambda_k \xleftrightarrow{\Lambda_{L_0} \cup D_0 \cup \dots \cup D_{t-1}} B_t \right) > 1 - \eta_t,$$

with the union interpreted as Λ_{L_0} when $t = 0$. The target lemma is applied in the graph induced by the path region in this display together with D_t . The wired source remains outside D_t , and the new lattice subbox is D_t . The lemma therefore advances the displayed inequality from t to $t + 1$ without allowing a path outside the next path region. Consequently B_N is reached with probability greater than $1 - \alpha$. The identities $L_N = Kr$ and $w_N \leq r$, which follow from (51), place B_N in the far face of the long box. All regions D_t lie in that box. Enlarging the source from Λ_k to Λ_m , $m \geq k$, proves (38) for $\gamma_{i,\sigma,K}$. $\square$

7.4 One corridor

The next lemma is the finite move used by the planar exploration. Its sets are recorded explicitly. For a centre c , direction (i, σ) , and scale r , put $c' = c + 20r\sigma e_i$ and

$$\begin{aligned} D &= c + \{x : -5r \leq \sigma x_i \leq 25r, |x_j| \leq 5r \ (j \neq i)\}, \\ E &= c + \{x : 5r + 1 \leq \sigma x_i \leq 25r, |x_j| \leq 5r \ (j \neq i)\}, \\ H &= c + \{x : 5r \leq \sigma x_i \leq 22r, |x_j| \leq 2r \ (j \neq i)\}. \end{aligned}$$

Lemma 7.5 (Corridor move). *Let $d \geq 3$ be an integer, and let $p \in (0, 1)$ satisfy $\theta_{\text{site}}(p) > 0$. For each $\alpha > 0$, there are $\beta > 0$ and an integer $r_0 \geq 1$ such that, for every integer $r \geq r_0$, the following holds. Let G be a finite site model for which D is a site subbox at parameter p . Let $o \notin D$ be forced open, and put*

$$A = V(G) \setminus E, \quad U = A \cup H.$$

Then

$$\mathbb{P}(o \xleftrightarrow{A} (c + \Lambda_{3r})) > 1 - \beta \implies \mathbb{P}(o \xleftrightarrow{U} (c' + \Lambda_{3r})) > 1 - \alpha. \quad (52)$$

Proof. Choose all error tolerances before choosing the scale. Apply Lemma 7.3 to the aspect-88 long-box list with output error α , and denote its input tolerance by η_d . For $0 \leq j < d$, working backwards, let η_j be the input tolerance for the quarter-face list with output error η_{j+1} . Let R be the maximum of the $d + 1$ resulting inflation radii, and put $\beta = \eta_0$.

We first shrink the cross-section near c . Put $h = r + \lceil r/2 \rceil$. In a fixed ordering of the d coordinates, let B_j be the box centred at c whose first j half-widths are $h + jR$ and whose remaining half-widths are $3r + jR$. Assume that $r > 100(d + 1)(R + 1)$. We verify that B_{j+1} is a target for B_j in $c + \Lambda_{5r}$. Let q be the coordinate whose width is reduced at this step, and write $h_1 = h + (j + 1)R$. For $v \in B_j^{+R}$, choose the face in coordinate q that points toward c and use the scale

$$\ell(v) = R + \max\{0, |v_q - c_q| - h_1\}.$$

Its q -coordinate lies in $[-h_1, h_1] + c_q$. In each transverse coordinate choose the orthant that points toward the corresponding coordinate of c . Every transverse coordinate on the face then has absolute displacement at most $\max\{|v_k - c_k|, \ell(v)\}$, which is within the corresponding half-width of B_{j+1} . Moreover,

$$R \leq \ell(v) \leq 3r - h + R < 2r.$$

The bounds on v , together with $r > 100(d + 1)(R + 1)$, place the entire cube of radius $\ell(v)$ in $c + \Lambda_{5r}$. Thus B_{j+1} is a target for B_j . After d applications of the target lemma in the graph induced by A , the source reaches B_d , whose half-widths are smaller than $2r$. This restriction is valid because $c + \Lambda_{5r} \subseteq A$.

For the final move use the finite list of long-box geometries from Lemma 7.4 with aspect ratio 88. Work in

$$D' = c + \{x : -2r \leq \sigma x_i \leq 22r, |x_j| \leq 2r \ (j \neq i)\}.$$

For $v \in B_d^{+R}$ choose

$$\ell(v) = \left\lfloor \frac{20r - \sigma(v_i - c_i)}{88} \right\rfloor. \quad (53)$$

When $r \geq r_0$, the long box at this scale lies in D' , and its far face lies in $c' + \Lambda_{2r}$. The lower bound containing the factor 100 absorbs the remainder in division by 88. Thus $c' + \Lambda_{2r}$ is a target for $(B_d, D', R, \mathcal{H}_{88})$. Pass now to the graph induced by $U = A \cup H$. The set D' is a site subbox in this graph, and the connection to B_d obtained in A remains valid. One more application of Lemma 7.3 reaches $c' + \Lambda_{2r}$. Enlarging the target to $c' + \Lambda_{3r}$ gives (52).

The induced graphs show that the first d steps use A and the last step adds only H . The backward choice of the $d + 1$ tolerances gives the asserted number β . $\square$

7.5 Planar exploration

We give the final finite-to-infinite step in a form that exposes the measurability needed for site variables. Embed $\mathbb{Z}^2$ in two coordinates of $\mathbb{Z}^d$ and, for $z \in \mathbb{Z}^2$, set

$$c_z = 20rz, \quad Q_z = c_z + \Lambda_{5r}, \quad M_z = c_z + \Lambda_{3r}.$$

For an oriented nearest-neighbour macro-edge $z \rightarrow w$, let i and σ be its coordinate and sign. Its open strip between the two endpoint cubes is

$$B_{z,w} = c_z + \{x : 5r + 1 \leq \sigma x_i \leq 15r - 1, |x_k| \leq 5r \ (k \neq i)\}.$$

The region exposed when w is examined from z is

$$E_{z,w} = B_{z,w} \cup Q_w = c_z + \{x : 5r + 1 \leq \sigma x_i \leq 25r, |x_k| \leq 5r \ (k \neq i)\}. \quad (54)$$

The full narrow corridor is

$$H_{z,w} = c_z + \{x : 5r \leq \sigma x_i \leq 22r, |x_k| \leq 2r \ (k \neq i)\}.$$

Choose $K \geq 20$, write $r = Ks$, and define the nested stubs

$$H_{z,w}^j = c_z + \{x : 5r \leq \sigma x_i \leq 5r + 10sj, |x_k| \leq 2r \ (k \neq i)\}, \quad 0 \leq j \leq K. \quad (55)$$

For $1 \leq j \leq K$, their outer faces are

$$F_{z,w}^j = c_z + \{x : \sigma x_i = 5r + 10sj, |x_k| \leq 2r \ (k \neq i)\}. \quad (56)$$

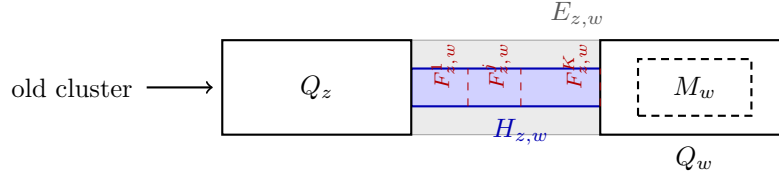


Figure 3: The regions used to examine the macro-edge $z \rightarrow w$, shown in the longitudinal coordinate. The broad edge region $E_{z,w}$ contains the target cube Q_w . The narrower corridor $H_{z,w}$ is revealed through nested faces. The examination stops at its first good face, so the unused part of the corridor remains fresh.

We now record the cover that controls every possible history. In the two planar coordinates, let

$$\text{Cell}(z) = c_z + [-10r, 10r]^2 \times [-5r, 5r]^{d-2}.$$

For the directed edge $z \rightarrow w$, define its stub zone by

$$Z_{z,w} = c_z + \{x : 10r \leq \sigma x_i \leq 15r - 10s, |x_k| \leq 2r \ (k \neq i)\}.$$

For a set $\mathcal{D}$ of determined macro-vertices, write

$$\text{Cover}(\mathcal{D}) = \bigcup_{z \in \mathcal{D}} \left(\text{Cell}(z) \cup \bigcup_{w: w \sim z} Z_{z,w} \right). \quad (57)$$

The next deterministic lemma is the freshness mechanism. Two lattice sets are called separated when they are disjoint and no nearest-neighbour edge has one endpoint in each set.

Lemma 7.6 (Cell separation). *Let $d \geq 3$, $K \geq 20$, and $s \geq 1$ be integers. Let $r = Ks$, and let $\mathcal{D} \subseteq \mathbb{Z}^2$ be a set of macro-vertices.*

- (C1) *If $z \notin \mathcal{D}$, then $\text{Cover}(\mathcal{D})$ and Q_z are separated.*
- (C2) *If $z, w \notin \mathcal{D}$ and $z \sim w$, then $\text{Cover}(\mathcal{D})$ and $E_{z,w}$ are separated.*
- (C3) *If $w \sim z$, then $E_{w,z} \subseteq \text{Cell}(w) \cup \text{Cell}(z)$. For $0 \leq j < K$ and $y \sim z$,*

$$H_{z,y}^j \subseteq \text{Cell}(z) \cup Z_{z,y}.$$

Proof. We compare the two planar coordinates. Distinct macro-centres differ by a multiple of $20r$ in at least one of them. A cell extends by $10r$ in each planar coordinate, whereas a cube extends by $5r$. A stub zone directed toward a neighbouring centre stops at level $15r - 10s$, while that neighbour's cube starts at level $15r$. These gaps are at least two lattice steps because $s \geq 1$. This proves (C1).

For (C2), first consider a cell or zone based at $u \in \mathcal{D}$. Since neither endpoint of the macro-edge zw belongs to $\mathcal{D}$, the vertex u differs from both endpoints. If the separating macro-coordinate is the direction of zw , then the preceding longitudinal bounds leave a gap of at least $5r$. Otherwise the transverse planar coordinate differs at both endpoints; the half-widths $10r$, $5r$, and $2r$ leave a gap of at least two lattice steps. The same comparison applies to a zone parallel to zw ; for a perpendicular zone, use the two distinct planar coordinates. This proves separation from every set in the union (57).

Finally, the strip $B_{w,z}$ is covered by the two planar cells on its ends, and $Q_z \subseteq \text{Cell}(z)$. This proves the first inclusion in (C3). If $j < K$, then

$$5r + 10sj \leq 15r - 10s.$$

The part of the stub below level $10r$ lies in $\text{Cell}(z)$, and its remaining part lies in $Z_{z,y}$. This proves the second inclusion. $\square$

Lemma 7.7 (Planar exploration). *Let $d \geq 3$ and $p \in (0, 1)$ satisfy $\theta_{\text{site}}(p) > 0$. For every $0 < \rho \leq 2^{-32}$, there are integers $K \geq 20$ and $s \geq 1$, with $r = Ks$, and a history-driven exploration of the macro-lattice with the following properties, conditional on all sites of Q_0 being open.*

- (P1) *Conditional on the complete past, each fresh macro-vertex examined by the exploration is declared occupied with probability at least $1 - \rho$.*
- (P2) *Every occupied macro-vertex z has a certificate a_z in the actual open cluster of the origin such that $\|a_z - c_z\|_\infty \leq 16r$.*
- (P3) *If $\mathcal{D}$ is the set of determined macro-vertices, then every inspected site belongs to $\text{Cover}(\mathcal{D})$.*
- (P4) *Before an undetermined vertex z is examined from an occupied neighbour, the old inspected set is separated from Q_z and from every $E_{z,y}$ with y also undetermined. Each site inspected for the first time therefore has, conditional on the complete past, its original independent Bernoulli law with parameter p .*

Proof. We first choose the parameters in the order in which the proof uses them. Apply Lemma 7.5 with output error $\rho/8$, and let $\delta_1 > 0$ be its input tolerance. Put $\delta_c = \min\{\delta_1, 1/2\}$. Apply Lemma 7.3 with output error δ_c , let $\delta_2 > 0$ be its input tolerance, and replace δ_2 by $\min\{\delta_2, 1\}$. Choose

$$K \geq 20, \quad (1 - \delta_2)^K + \frac{\rho}{8} \leq \frac{\rho}{4}. \quad (58)$$

Apply the target lemma to the finite list of long geometries of aspect $2K$, and let R be the resulting inflation radius. Choose s so large that $s \geq 2R$, $r = Ks$ is beyond the corridor threshold, and the long geometries of aspect 6 are hittable at scale $3r$ with error δ_c . This completes the parameter choices.

We now define the invariant propagated by the exploration. A history records a finite set $\mathcal{E}$ of inspected sites and its open subset ξ , together with disjoint sets of occupied and blocked macro-vertices. Their union $\mathcal{D}$ is the determined set. Suppose that w is occupied, that z is

undetermined, and that the current oriented macro-edge is $w \rightarrow z$. The history is valid when $\mathcal{E} \subseteq \text{Cover}(\mathcal{D})$ and, in the finite site model obtained by pinning $\mathcal{E}$ to ξ , the conditional probability of

$$0 \xleftrightarrow{\mathcal{E} \cup E_{w,z}} M_z \quad (59)$$

is greater than $1 - \delta_c$. All unpinned sites in the displayed region retain parameter p . Initially, 0 is the sole occupied macro-vertex, $\mathcal{E} = Q_0$, and all sites of Q_0 are pinned open. For an edge $0 \rightarrow z$, place the aspect-6 long geometry at scale $3r$ along that edge. Its box lies in $Q_0 \cup E_{0,z}$, and its far face lies in M_z . Long-box hitting gives a connection from Λ_{3r} to that face with probability greater than $1 - \delta_c$. Conditioning on the increasing event that every site of Q_0 is open can only increase this probability, by Harris' inequality. Since the conditioning joins 0 to every site of Λ_{3r} , we obtain (59) for each of the four macro-edges leaving the origin. The initial cover condition follows from $Q_0 \subseteq \text{Cell}(0)$.

Assume that a valid history with current edge $w \rightarrow z$ has been reached. Let X be the set of neighbours y of z whose macro-cubes have not been determined. First reveal the previously uninspected sites of $E_{w,z}$. For each $y \in X$, reveal the nested stub one layer at a time, stopping as soon as a good level is found. At level j , pin the sites of

$$\mathcal{E} \cup E_{w,z} \cup H_{z,y}^j$$

to their observed states. Call level j good if the resulting conditional probability of

$$0 \xleftrightarrow{\mathcal{E} \cup E_{w,z} \cup H_{z,y}^j \cup E_{z,y}} M_y \quad (60)$$

is greater than $1 - \delta_c$. Stop at the first good level, or at $K - 1$ if no good level exists. Declare z occupied exactly when a good level was found for every $y \in X$; otherwise declare it blocked. In both cases add $E_{w,z}$ and the selected stubs to $\mathcal{E}$, and add z to the determined set. Lemma 7.6(C3) preserves the cover condition (P3).

On an occupied examination, (60) supplies the required lower bound for each outgoing edge $z \rightarrow y$. Outside the already pinned part $E_{w,z}$, stubs in distinct directions are disjoint. The event tested for direction y does not use the other stubs, so conditioning on their states does not change its probability. The new allowed region contains the region used in that test. Hence its connection probability is still greater than $1 - \delta_c$, and the invariant propagates to every outgoing candidate.

We next bound the probability that one direction has no good level. In the full corridor from z to y , consider the conditional model before the examination of z . Lemma 7.6 separates the old read set from $Q_z \cup E_{z,y}$. Moreover, $y \neq w$, so $E_{w,z}$ is disjoint from $E_{z,y}$. Restrict the pinned model to the graph induced by $\mathcal{E} \cup E_{w,z} \cup E_{z,y}$. In this graph, $Q_z \cup E_{z,y}$ is the site subbox D from the corridor lemma, $E_{z,y}$ is its far region E , and $\mathcal{E} \cup E_{w,z}$ is the region A . Thus (59) is exactly the hypothesis of Lemma 7.5. That lemma gives

$$\mathbb{P} \left(0 \xleftrightarrow{\mathcal{E} \cup E_{w,z} \cup H_{z,y}} M_y \mid \text{past} \right) > 1 - \frac{\rho}{8}. \quad (61)$$

For $0 \leq j < K$, define

$$A_j = \left\{ 0 \xleftrightarrow{\mathcal{E} \cup E_{w,z} \cup H_{z,y}^{j+1}} F_{z,y}^{j+1} \right\},$$

and let B_j be the event that level j is not good. If B_j occurs, then the conditional probability of A_j , given the past and all sites in $E_{w,z} \cup H_{z,y}^j$, is at most $1 - \delta_2$. To verify this implication, use the uninspected subbox

$$D_j = c_z + \{x : 5r + 10sj + 1 \leq \sigma x_i \leq 25r, |x_k| \leq 5r \ (k \neq i)\}.$$

The separation lemma and the disjointness of the directional stubs imply that every site of D_j is still an independent parameter- p site. After pinning through $H_{z,y}^j$, restrict the graph to the pinned source region together with D_j . This makes D_j a site subbox whose lower boundary is adjacent to the end of $H_{z,y}^j$. We verify that M_y is a target for the source face $F_{z,y}^{j+1}$ in D_j , using the aspect- $2K$ long geometry. Write

$$a_j = 5r + 10s(j+1), \quad \ell_j = \left\lfloor \frac{20r - a_j}{2K} \right\rfloor.$$

For a point v in the R -enlargement of $F_{z,y}^{j+1}$, we have

$$a_j - R \leq \sigma(v - c_z)_i \leq a_j + R, \quad |(v - c_z)_k| \leq 2r + R \quad (k \neq i).$$

The assumptions $0 \leq j < K$, $r = Ks$, $K \geq 20$, and $2R \leq s$ imply

$$R \leq \ell_j \leq \frac{15s}{2}, \quad \ell_j + R \leq 8s < 10s.$$

The long box based at v therefore starts after the lower face of D_j . Its forward coordinate is at most $20r + R < 25r$, and its transverse coordinates have absolute displacement at most $2r + R + \ell_j \leq 3r$. If $q_j = 2K\ell_j$, then

$$20r - R - (2K - 1) \leq \sigma(v - c_z)_i + q_j \leq 20r + R.$$

This interval lies in $[17r, 23r]$, so the far face lies in M_y ; the full long box lies in D_j . We have proved the target condition uniformly over the R -enlargement of $F_{z,y}^{j+1}$. Consequently, a conditional probability larger than $1 - \delta_2$ for A_j would let Lemma 7.3 extend the connection to M_y with probability greater than $1 - \delta_c$, contrary to B_j . Iterated conditioning therefore gives

$$\mathbb{P} \left(\bigcap_{j=0}^{K-1} (A_j \cap B_j) \mid \text{past} \right) \leq (1 - \delta_2)^K. \quad (62)$$

By Lemma 7.6(C2), no site of the old read set is adjacent to $E_{z,y}$. The region $E_{w,z}$ reaches only the source side of $F_{z,y}^1$. Every path counted in (61) therefore starts on the source side of $F_{z,y}^1$ and ends in M_y , beyond $F_{z,y}^K$. Along a nearest-neighbour path the signed i -coordinate changes by at most one at each step. The path consequently meets $F_{z,y}^1, \dots, F_{z,y}^K$ in this order, and its prefix to $F_{z,y}^{j+1}$ is contained in $\mathcal{E} \cup E_{w,z} \cup H_{z,y}^{j+1}$. Hence the event that all K levels are bad is contained in the failure event from (61) together with the event in (62). Its conditional probability is at most $\rho/4$ by (58). At most four directions belong to X , so a union bound proves (P1).

We verify freshness by induction over examinations. The initial inspected set Q_0 lies in $\text{Cover}(\{0\})$. Suppose the cover condition holds before z is examined. Lemma 7.6(C1)–(C2) separates the old inspected set from Q_z and from each outgoing edge region whose endpoints are undetermined. We inspect only sites outside $\mathcal{E}$; product independence gives their conditional Bernoulli laws. After the examination, Lemma 7.6(C3) places the incoming edge region and every selected stub in the cover of $\mathcal{D} \cup \{z\}$. This proves (P3) and (P4) simultaneously for every history, including histories in which the examination fails.

Finally, consider an occupied macro-vertex $z \neq 0$ and the valid history by which it was reached from a neighbour w . The probability in (59) is positive, so there exists a site path in the pinned finite model from the origin to M_z . Choose such a path in a configuration of positive conditional weight. Since $M_z \subseteq E_{w,z}$ and $0 \notin E_{w,z}$, the path enters $E_{w,z}$. Let a_z be its last vertex before the first entrance. The prefix from 0 to a_z lies in $\mathcal{E}$. Every open site on that prefix is therefore pinned

open by the recorded, actual configuration. Thus a_z belongs to the actual open cluster of 0. The coordinate bounds in (54) give $\|a_z - c_z\|_\infty \leq 16r$. For $z = 0$, take $a_0 = 0$. This proves (P2) and completes the construction. $\square$

Proposition 7.8 (Same-parameter slab reduction). *Let $d \geq 3$ and $p \in (0, 1)$. If $\theta_{\text{site}}(p) > 0$, then there is $r \geq 1$ such that Bernoulli site percolation with parameter p percolates in*

$$\mathbb{Z}^2 \times [-5r, 5r]^{d-2}.$$

Proof. Choose ρ so that $1 - \rho$ is larger than the critical probability for site percolation on $\mathbb{Z}^2$, and use Lemma 7.7. Run the usual cluster exploration from the origin: at each step, examine the first undetermined neighbour of the retained cluster in a fixed deterministic order. Every retained vertex is occupied, so each requested examination has a valid occupied predecessor. Property (P1) gives its conditional success probability $q \geq 1 - \rho$.

Attach an independent uniform random variable to each macro-vertex. Retain a successful examination exactly when this variable is at most $(1 - \rho)/q$. Conditional on the complete past, the retained indicator has probability $1 - \rho$. It is also independent of the retained indicators already exposed. Induction over the exploration steps therefore identifies the retained cluster with the origin cluster of Bernoulli site percolation on $\mathbb{Z}^2$ at parameter $1 - \rho$, conditional on the origin being open. Since $1 - \rho$ is supercritical, this cluster is infinite with positive probability. Every retained vertex belongs to the occupied macro-process.

On this event there are infinitely many occupied macro-vertices z . Property (P2) assigns to each of them a site a_z in the actual cluster of the origin and within $16r$ of c_z . A fixed lattice site lies within $16r$ of only finitely many of the centres $c_z = 20rz$. The set $\{a_z : z \text{ is occupied}\}$ is therefore infinite. Property (P3) puts the certificates and their paths inside the displayed slab. Thus the origin has an infinite open cluster in that slab.

We initially forced the finite set Q_0 open. This event has probability $p^{|Q_0|} > 0$. After integrating out the auxiliary uniform variables, the preceding construction still gives a positive conditional probability of slab percolation. Its intersection with the event that Q_0 is open has positive unconditioned probability. The unconditioned site model therefore percolates in the slab. $\square$

8 Criticality

We now complete the proof. We use the following established half-space theorem in its site-percolation form: for $d \geq 3$, the critical probability of a half-space equals $p_c^{\text{site}}(\mathbb{Z}^d)$, and the half-space does not percolate at that parameter. This is the site case of the theorem of Barsky, Grimmett, and Newman [2].

Proof of Theorem 1.1. For $d = 2$, absence of an infinite cluster at the critical point is the classical planar site-percolation theorem; see Russo [8].

Let $d \geq 3$, and abbreviate $p = p_c^{\text{site}}(\mathbb{Z}^d)$. Standard degree and contour bounds give $0 < p < 1$. Suppose for a contradiction that $\theta_{\text{site}}(p) > 0$. Proposition 7.8 gives a finite-width slab that percolates at the same parameter p . Translate the slab so that it lies in the half-space

$$\{x \in \mathbb{Z}^d : x_d \geq 0\}.$$

Percolation is monotone under enlargement of the underlying graph, so the half-space also percolates at p . This contradicts the site half-space theorem. Therefore $\theta_{\text{site}}(p) = 0$. $\square$

9 Formalization targets

The proof separates into finite and lattice modules. A formalization should start with the finite part, because the lattice reduction is useful only after Theorem 5.4 has been checked.

The finite dependency order is as follows.

1. Define finite labelled hypergraphs, induced deletion, hyperedge clusters, set clusters, and product weights.
2. Prove the exploration identity in Lemma 2.2 and exact conditioning on a cluster.
3. Prove the one-cluster and two-cluster inequalities, Lemmas 3.1 and 3.2.
4. Define anchored labels and prove Lemma 4.2 by strong induction on the number of vertices. The formal statement must quantify over the label type and anchor map before the induction begins.
5. Port the conditioned covariance hierarchy. Only its two-source averaging step should invoke the labelled lemma; the level operators and the surplus peeling are algebraic.
6. Derive Theorems 5.1 and 5.4, then formalize the site-to-hyperedge representation and Corollary 6.1.

The decisive regression test is a pair of distinct incident hyperedges with the same outside anchor, as in Figure 2. A formal proof that pushes exposed states directly to vertex traces will incorrectly replace an inclusion by an equality. The new labels must remain distinct through the union and intersection in (13) and (14).

For the lattice part, duplicate the bond-percolation target and corridor interfaces with site configurations. Conditioning pins vertices rather than edges. In the many-contact estimate, close the at most $2dN$ fresh interior neighbor sites. In the seed estimate, use disjoint finite vertex seeds. Formalize Lemma 7.6 before the planar exploration; its cover invariant is what keeps each outgoing site subbox fresh after an arbitrary exploration history.

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